

SECTION 1: PROCESS DESCRIPTION

My project is about creating a business rules model to automate the unit enrolment process at Macquarie University. The aim is to ensure that the students who meet various academic and administrative conditions, such as GPA, prerequisites, seat availability, year, and fee payment, are allowed to enrol in a unit. Final year students may be granted exceptions if other conditions are reasonably met.

The rules are implemented in the JDM Go Rules Editor and executed via the ZEN rule engine. The model is primarily inspired by my unit enrolment experiences at Macquarie University, involving one request node, one decision table, and one response node. The user will input their information in the form of data structures through the simulator. This information will go through the request node into the decision table, which will provide a decision based on the business rules, and forward it to the response node. Some cases have been formulated and tested to understand how this model responds in various scenarios.

I believe this will be a good example for a business rules model because it is a process that requires significant attention to detail. Many students and administrative staff go through the nitty-gritty of this process. Automating the unit enrolment process at Macquarie University will significantly enhance the administrative and operational efficiency of the university.

Also, this will reduce the burden on the administrative and IT staff regarding the unit enrolment process. This will provide them space to further improve the processes at the university, along with other relevant departments. Overall, it will enhance the image of Macquarie University among the current students and prospective future students, who want to study and improve their future with the university's student-centric approach (You raised to the power of us).

SECTION 2: DATA STRUCTURES

- `student.gpa` (number): Student's Grade Point Average.
- `student.year` (string): Student's current academic year ("First Year", "Final Year", etc).
- `prerequisites.completed` (boolean): Indicates if the student completed the prerequisites or not
- `seats.available` (boolean): Whether the unit still has vacant seats.
- `paid.fees` (boolean): Whether the student has paid their tuition fees.
- `enrolment.approved` (Boolean): This is the output describing enrolment status (approved or not)

SECTION 3: BUSINESS RULE STATEMENTS

1. If GPA is equal to 3 or higher, prerequisites are completed, seats are available, and fees are paid, then enrolment is approved.
2. If GPA is equal to 3 or higher, prerequisites are completed, seats are available, and fees have not been paid, then enrolment is not approved.
3. Enrolment for students with a GPA less than 3 is not approved under any condition.
4. If the seats are unavailable, enrolment is not approved, regardless of other conditions.
5. If GPA is equal to 3 or higher, seats are available, the student is in the final year, and fees are paid, then the incomplete prerequisites are waived off and enrolment is approved.
6. If fees are not paid, enrolment is denied under all circumstances

SECTION 4: TEST CASES

TEST CASE 1: Seats not available (Expected Result: Enrolment Not Approved, Actual Result: Enrolment Not Approved)

```
{
  "student": {
    "gpa": 3.2,
    "year": "First Year"
  },
  "prerequisites": {
    "completed": true
  },
  "seats": {
    "available": false
  },
  "paid": {
    "fees": false
  }
}
```

TEST CASE 2: Final Year Exception (Expected Result: Enrolment Approved, Actual Result: Enrolment Approved)

```
{
  "student": {
    "gpa": 3.4,
    "year": "Final Year"
  },
  "prerequisites": {
    "completed": false
  },
  "seats": {
    "available": true
  },
  "paid": {
    "fees": true
  }
}
```

TEST CASE 3: GPA Too Low (Expected Result: Enrolment Not Approved, Actual Result: Enrolment Not Approved)

```
{
  "student": {
    "gpa": 2.5,
    "year": "Second Year"
  },
  "prerequisites": {
    "completed": true
  },
}
```

```
"seats": {  
  "available": true  
},  
"paid": {  
  "fees": true  
}  
}
```

TEST CASE 4: Fees Not Paid (Expected Result: Enrolment Not Approved, Actual Result: Enrolment Not Approved)

```
{  
  "student": {  
    "gpa": 3.7,  
    "year": "Final Year"  
  },  
  "prerequisites": {  
    "completed": true  
  },  
  "seats": {  
    "available": true  
  },  
  "paid": {  
    "fees": false  
  }  
}
```